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TRANSITIONS BETWEEN 2:1 AND 1:1 RESPONSES IN CARDIAC MUSCLE INDUCED BY ADDED STIMULI

Sonya Bahar, G. Martin Hall, Robert A. Oliver, Wanda Krassowska and Daniel J. Gauthier
Department of Physics, Department of Biomedical Engineering, and Center for Nonlinear and Complex Systems, Duke University, Box 90305, Durham, NC 27708
bahar@phy.duke.edu

ABSTRACT

We demonstrate experimentally that transitions can be induced between stable coexisting response patterns in bullfrog ventricular myocardium by means of a single pulse injected into the stimulus train. We find that two distinct ranges of stimulus timings will elicit transitions from a 1:1 state, in which one stimulus elicits one response, and a 2:1 state, in which every other stimulus elicits a response. We also observe two distinct types of 2:1 → 1:1 transition. These transitions are differentiated by their transient behavior while settling from one stable state into another. We characterize these transitions, and the ranges of added stimulus timings which produce them, both experimentally and with two simple models of cardiac dynamics.

INTRODUCTION

In small pieces of periodically paced bullfrog myocardium, two stable response patterns, 1:1 and 2:1, are observed to coexist over a wide range of pacing frequencies; the system is said to exhibit *bistability* [1]. Chaos control has recently been explored as a method of modulating cardiac dynamics using small stimuli rather than large shocks [2,3]. However, chaos control protocols are applicable to stablilization of unstable periodic states, not to modifiying a system between two stable states. A single extra stimulus (S_2) injected into the train of paces can induce transitions between bistable states. The extra stimulus is of the same current amplitude and duration as the regular pacing applied to the tissue: only the timing is changed.

METHODS

Small pieces of periodically paced bullfrog *(Rana catesbeiana)* ventricular myocardium, quiescent in absence of an applied stimulus, are paced with 4 ms square pulses generated by a computer-controlled Grass S48 stimulator. Periodic pacing is applied through two 51 μm tungsten wires set ~1-2 mm apart on the tissue surface. The amplitude of the applied stimulus is typically twice the excitation threshold. Under these conditions simple dynamical responses (typically 1:1, 2:1 and 2:2 states) are observed. The transmembrane potential is measured within 1-2 mm of the stimulus electrodes using a glass micropipette. Before starting data collection, the tissue is paced at ~1000 ms intervals for about 20 min. At each interstimulus interval we record the tissue response for up to 10 seconds after discarding the response to the last 5-10 stimuli in order to eliminate transients. Once the range of pacing intervals over which both 1:1 and 2:1 responses occur has been determined, we bias the tissue onto one behavior by sweeping the pacing interval up and down throught the bistability window.

RESULTS

Of the 12 animals studied, 9 exhibited coexisting 1:1 and 2:1 responses. Transitions were successfully induced between bistable states in multiple experiments from five of these animals. Experiments from three of the five animals in which transitions were observed showed 1:1 → 2:1 transitions. Separate experiments from all five animals showed 2:1 → 1:1 transitions. We find two different types of 2:1 → 1:1 transition, and also two different types of 1:1 → 2:1 transition. These transition types differ in their transient response to the S_2 stimulus as the tissue settles to a new stable state. A simple iterative mapping model and an ionic model account qualitatively for the observed 2:1 → 1:1 transitions and for one type of 1:1 → 2:1 transition.

DISCUSSION

These results may have implications for the control of cardiac rhythms. In the clinic, the sino-atrial node sets the lower limit on the frequency of stimuli delivered to the atrium. Additional stimuli can be applied, but individual SA node impulses cannot be *removed*. This is analogous to our experimental case of cardiac tissue driven by an applied train of stimuli. In a realistic experimental model, we cannot subtract stimuli from the train in order to modulate the dynamics. Therefore, the modification of cardiac dynamics by injected stimuli may be more directly applicable to clinical settings than control techniques [2,3] which require the variation of the interstimulus interval as a control parameter.

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